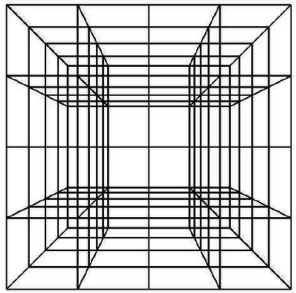
Comparison of neural representation and compressors

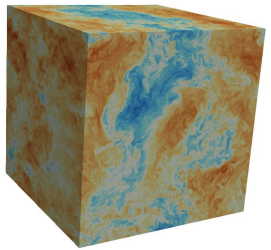
Hanqi Guo

Data Representation: From Grids/Points to Neural Fields

Standard scientific data representations

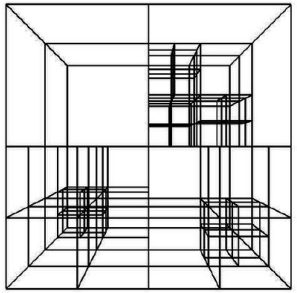
- regular grids[1]

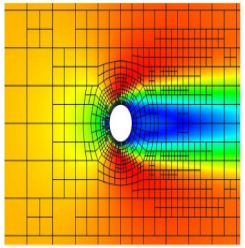




JHTDB turbulence[2]

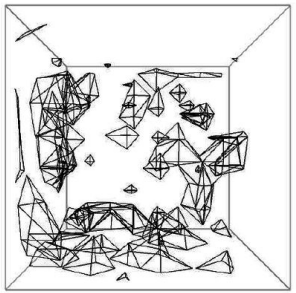
- adaptive/multi-resolution grids[1]

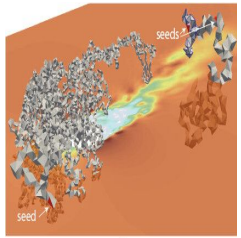




Rek5000 CFD[3]

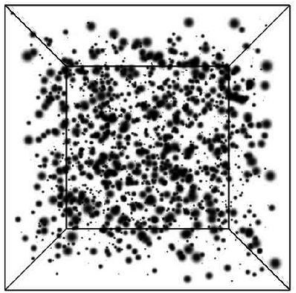
- unstructured meshes[1]

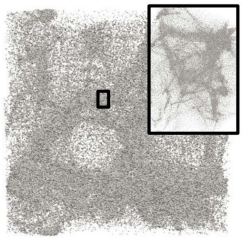




LES CFD[4]

- scattered particles[1]





Nyx cosmology[5]

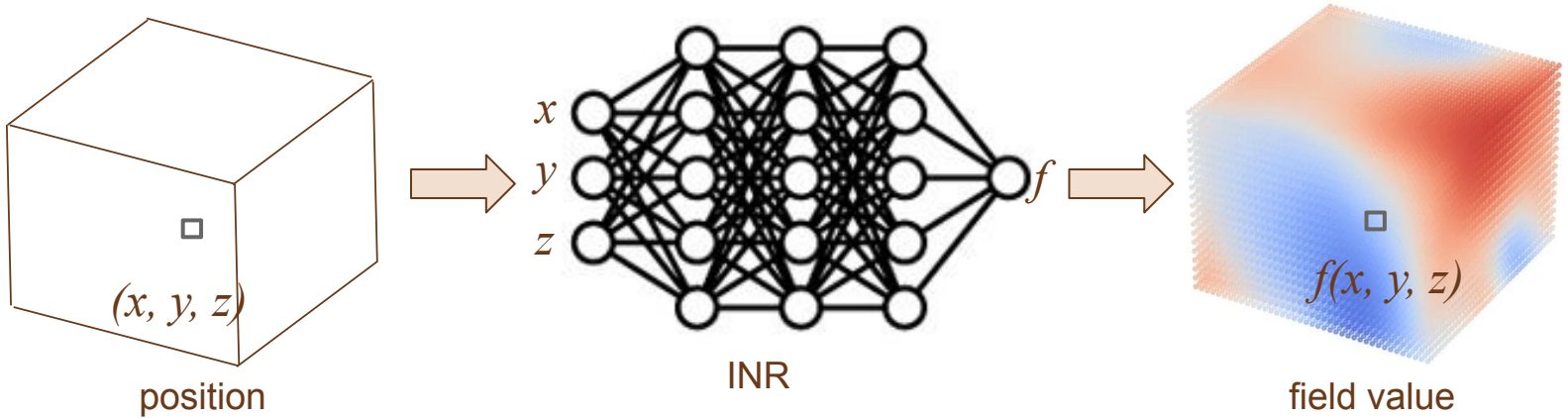
Drawback: cubic memory scaling

spatial resolution	memory footprint (FP32)
512^3	512 MB
$2{,}048^3$	32 GB
$8{,}192^3$	2 TB

[1] Hassan, A., & Fluke, C. J. (2011). Scientific visualization in astronomy: Towards the petascale astronomy era. *Publications of the Astronomical Society of Australia*, *28*(2), 150-170. [2] Johns Hopkins Turbulence Database, https://turbulence.idies.jhu.edu/datasets [3] Offermans, N., Peplinski, A., Marin, O., & Schlatter, P. (2020). Adaptive mesh refinement for steady flows in Nek5000. *Computers & Fluids*, *197*, 104352. [4] Ren, C., Liang, X., & Guo, H. (2024, June). A Prediction-Traversal Approach for Compressing Scientific Data on Unstructured Meshes with Bounded Error. In *Computer Graphics Forum* (Vol. 43, No. 3, p. e15097). [5] Ren, C., Di, S., Zhang, L., Zhao, K., & Guo, H. (2024). An error-bounded lossy compression method with bit-adaptive quantization for particle data. *arXiv preprint arXiv:2404.02826*.

Data Representation: From Grids/Points to Neural Fields

Implicit neural representations (INRs)



- Compact: stores neuron weights instead of values at fixed data points
- Continuous: queries any spatial location
- Differentiable: provides gradient by the network itself

Evolution Timeline of INR

2019

Shapes as coordinate MLPs

Continuous 3D geometry representation via dense coordinate MLPs

- DeepSDF
- Occupancy networks

2020

High-frequency fidelity

Positional encoding and sinusoidal activations

- NeRF
- SIREN

2021

First scientific use

Volume compression and rendering

- neurcomp
- fv-SRN

2022

Speed

Learned grid encodings with a tiny MLP; fitting in seconds to minutes

- Instant NGP
- TensoRF

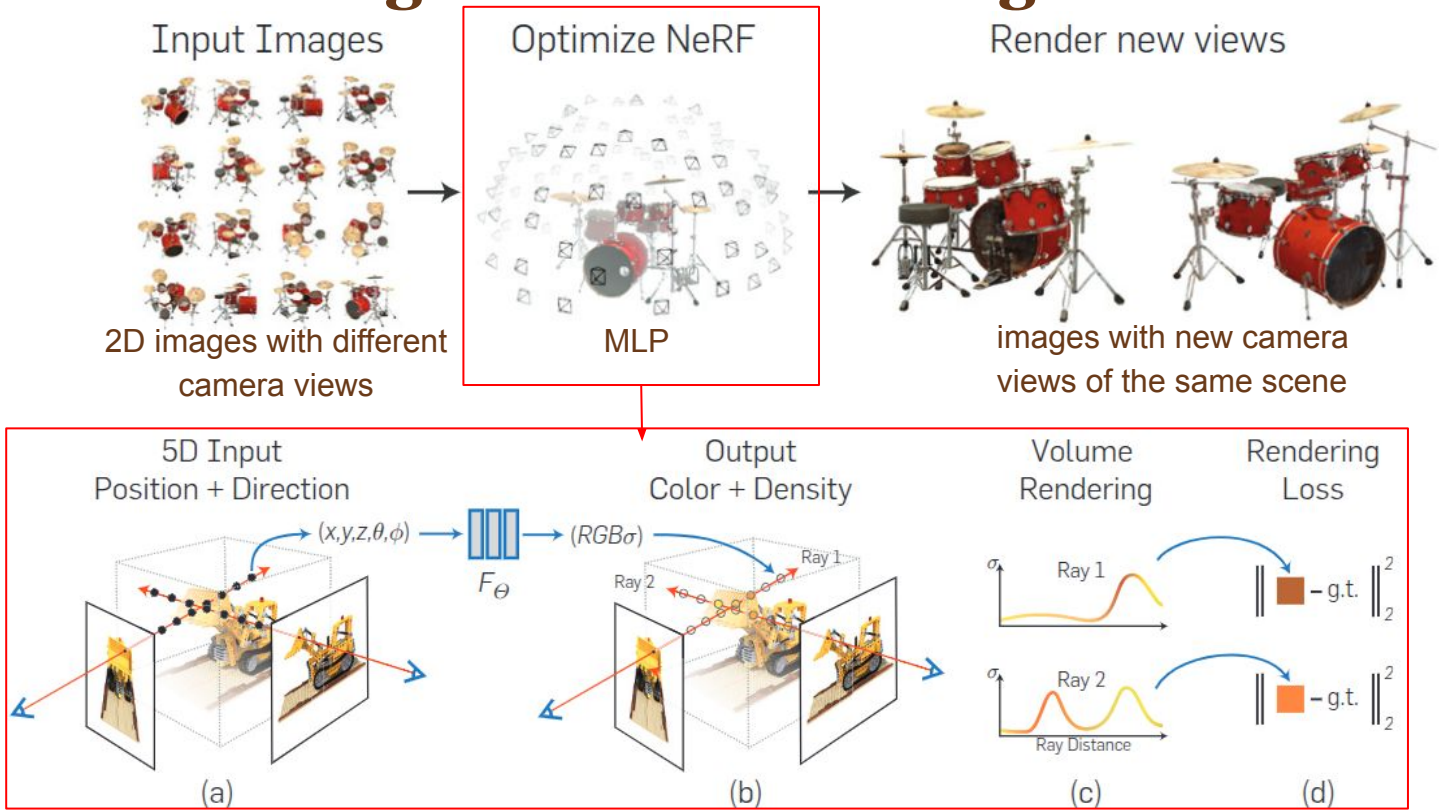
2023-present

Time-varying and ensemble data

Scaling beyond single volumes, toward general and explicit representations

- CoordNet, ECNR (time-varying compression)
- Meta-INR, MoE-INR (pretrain and adapt)
- 3D Gaussian Splatting, iVR-GS (explicit, editable scenes)

NeRF: rendering a scene from images



Mildenhall, Ben, et al. "NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis." Computer Vision – ECCV 2020, edited by Andrea Vedaldi et al., Springer, 2020, pp. 405–421,

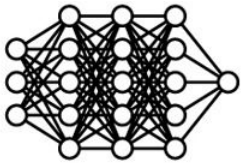
SIREN: Implicit Neural Representations with Periodic Activation Functions

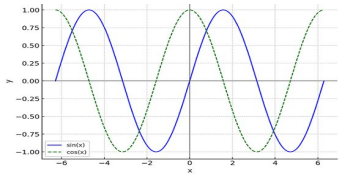
- SIREN features sine as a periodic activation function in MLPs:

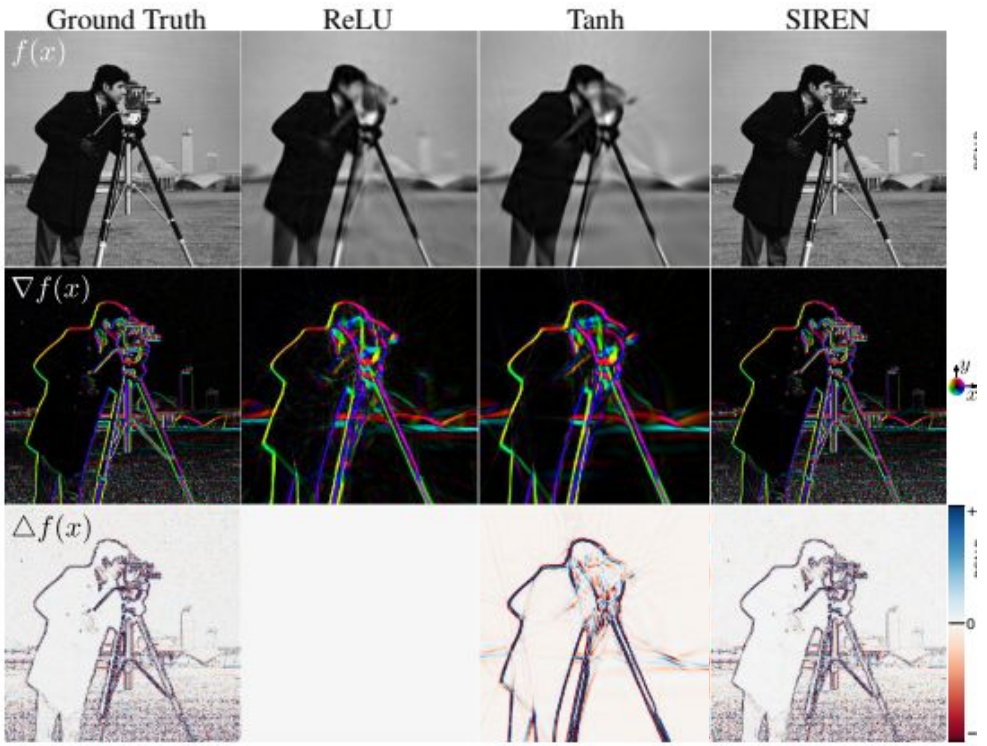
$$\Phi(\mathbf{x}) = \mathbf{W}_n(\phi_{n-1} \circ \phi_{n-2} \circ \ldots \circ \phi_0)(\mathbf{x}) + \mathbf{b}_n$$

$$\phi_i(\mathbf{x}_i) = \sin(\mathbf{W}_i\mathbf{x}_i + \mathbf{b}_i).$$

- Benefits: compact, infinitely differentiable, and ability to represent high-frequency details



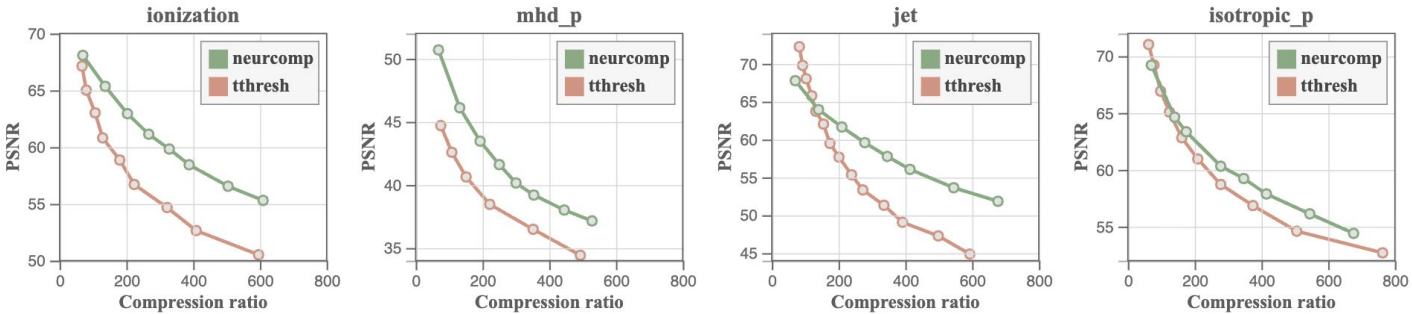


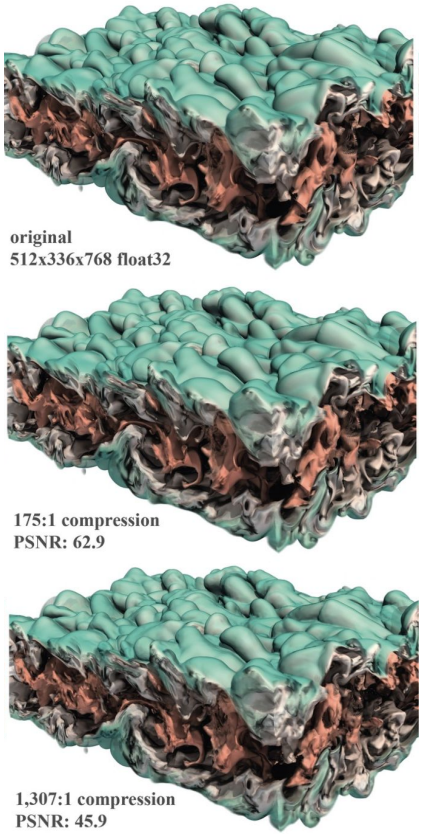


Sitzmann, Vincent, et al. "Implicit Neural Representations with Periodic Activation Functions." Advances in Neural Information Processing Systems, edited by Hugo Larochelle et al., vol. 33, Curran Associates, 2020, pp. 7462–7473. NeurIPS Proceedings,

NeurComp

- Using SIREN for representing 3D volumetric data
- Quantization is used to improve the compression ratio, measured by the original size vs. bytes for storing network weights

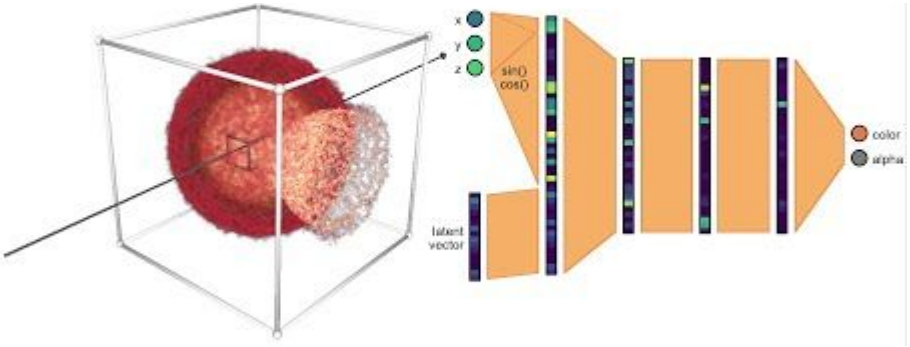


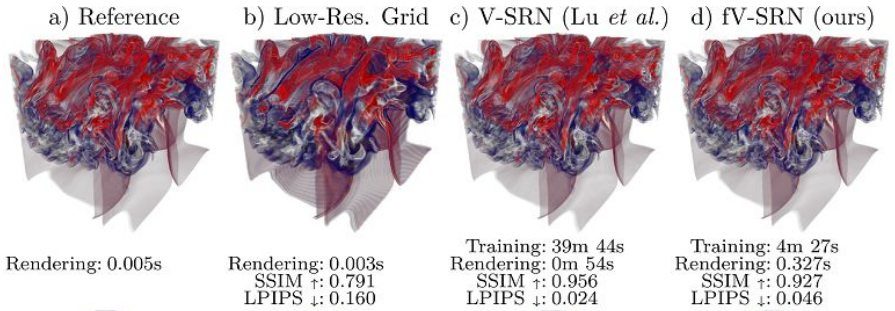


Lu, Yuzhe, et al. "Compressive Neural Representations of Volumetric Scalar Fields." Computer Graphics Forum, vol. 40, no. 3, 2021, pp. 135–146, doi.org.

fV-SRN: Fast Neural Representations for Direct Volume Rendering

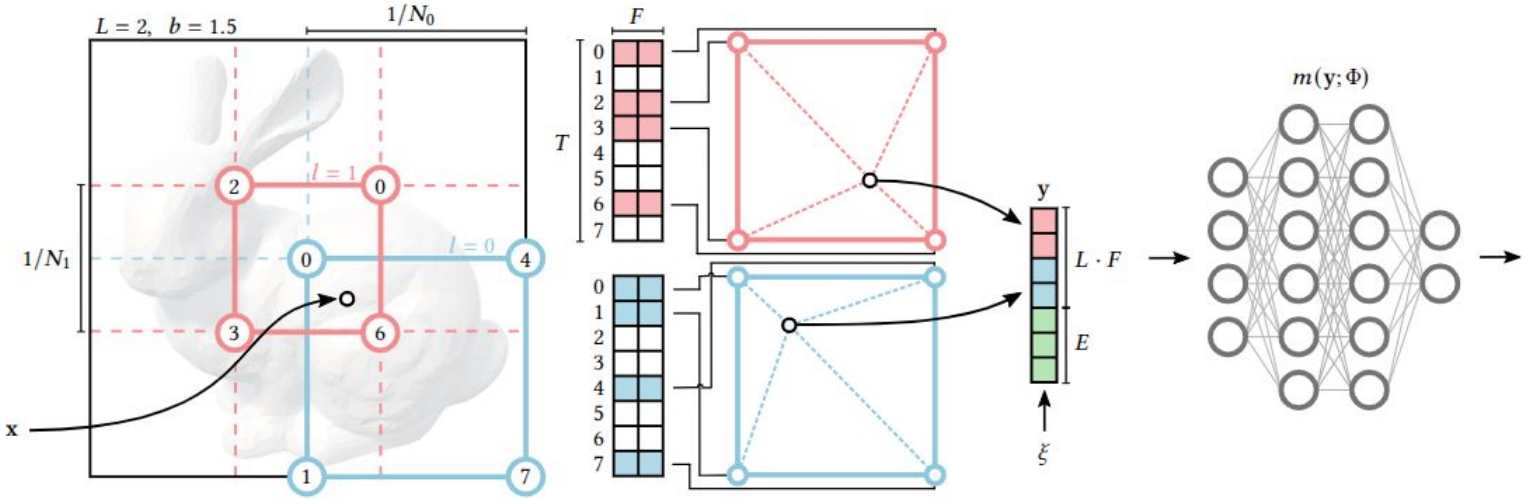
- fV-SRN introduced a volumetric lated grid
- Spatial-dependent latent vectors are fed to the MLP in conjunction with spatial coordinates
- Hardware accelerated implementation optimized for direct volume rendering





Weiss, Sebastian, et al. "Fast Neural Representations for Direct Volume Rendering." arXiv, 14 Mar. 2022, doi.org.

NGP with a multiresolution hash encoding



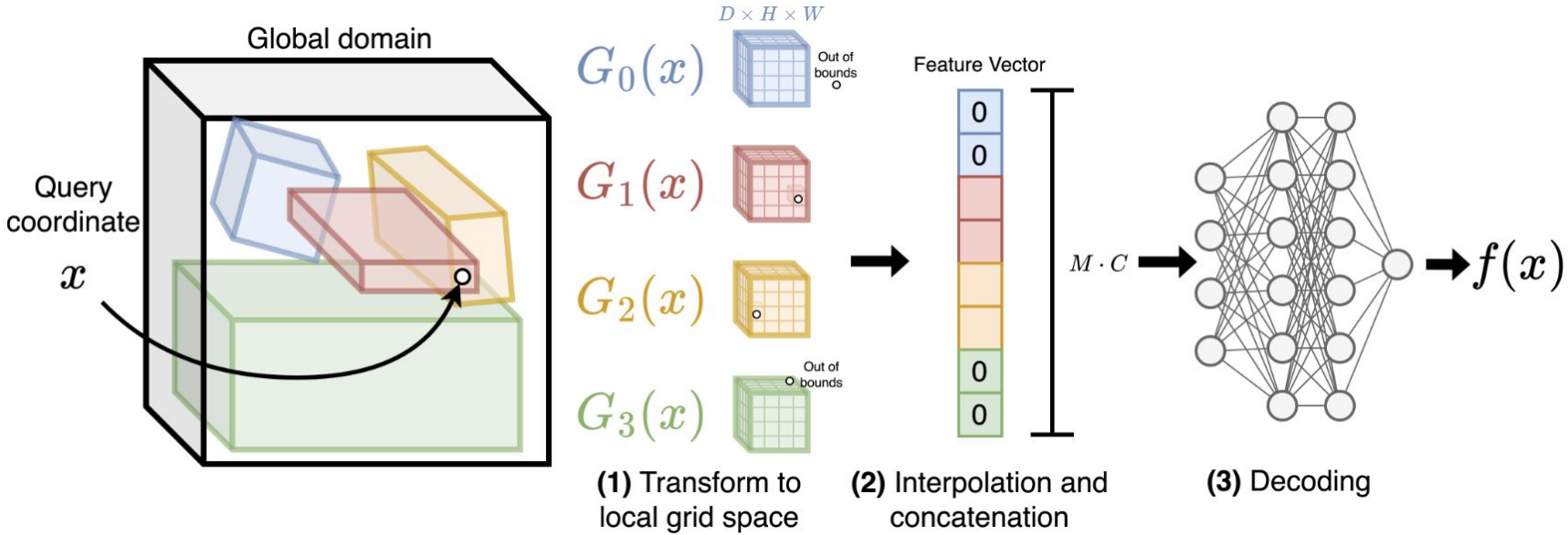
encode position by multi-resolution hash encoding

fully fused network -> single GPU kernel, less memory overhead

Müller, Thomas, et al. "Instant Neural Graphics Primitives with a Multiresolution Hash Encoding." ACM Transactions on Graphics, vol. 41, no. 4, July 2022, pp. 102:1–102:15. ACM Digital Library, https://doi.org/10.1145/3528223.3530127.

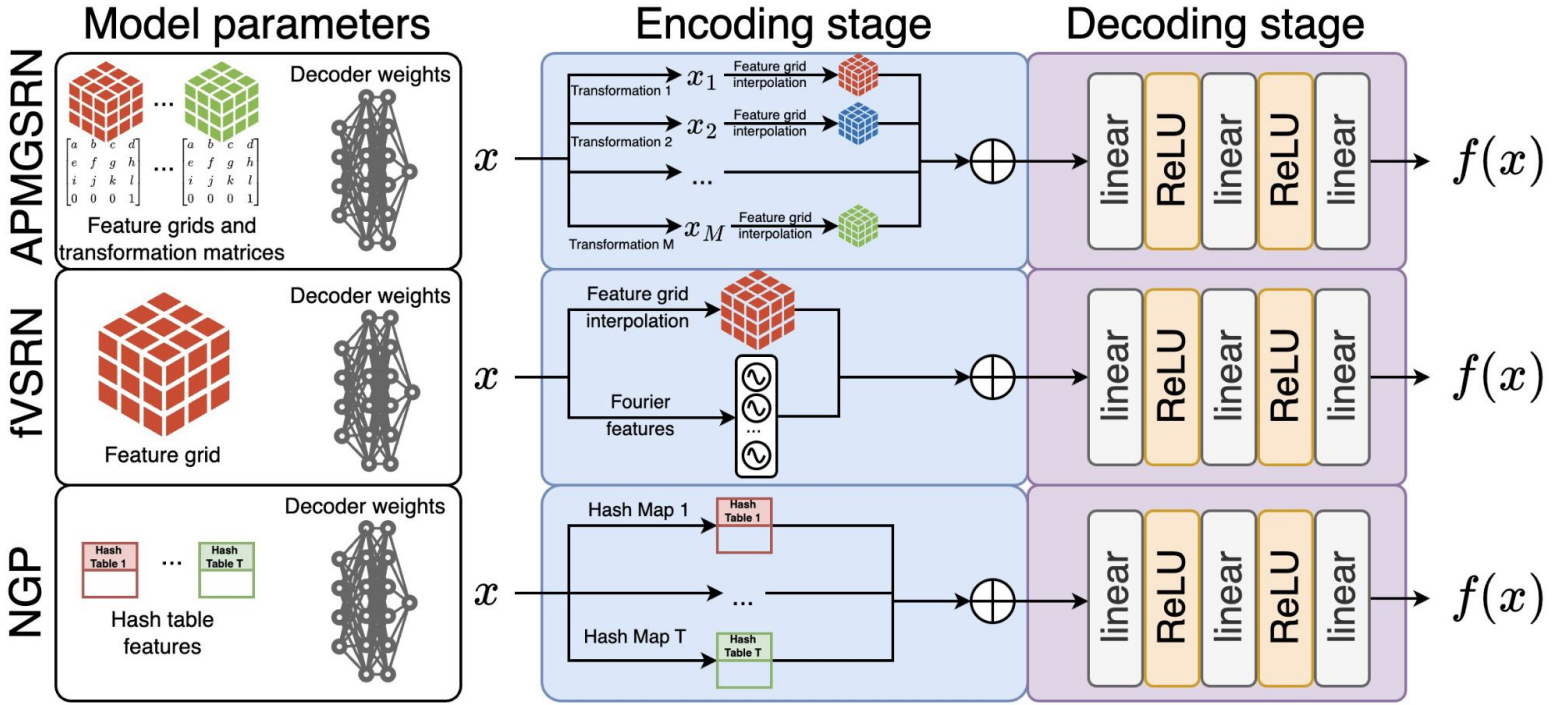
APMGSRN: Adaptively Placed Multi-Grid SRN

- Another multi-grid approach that adaptively allocate feature grid based on spatial distributions



Skylar Wurster, Tianyu Xiong, Han-Wei Shen, Hanqi Guo, and Tom Peterka, “Adaptively Placed Multi-Grid Scene Representation Networks for Large-Scale Data Visualization.” IEEE Transactions on Visualization and Computer Graphics (Proc. IEEE VIS 2023), 30(1):965-974, 2024.

Comparisons of fVSRN, NGP, and APMGSRN

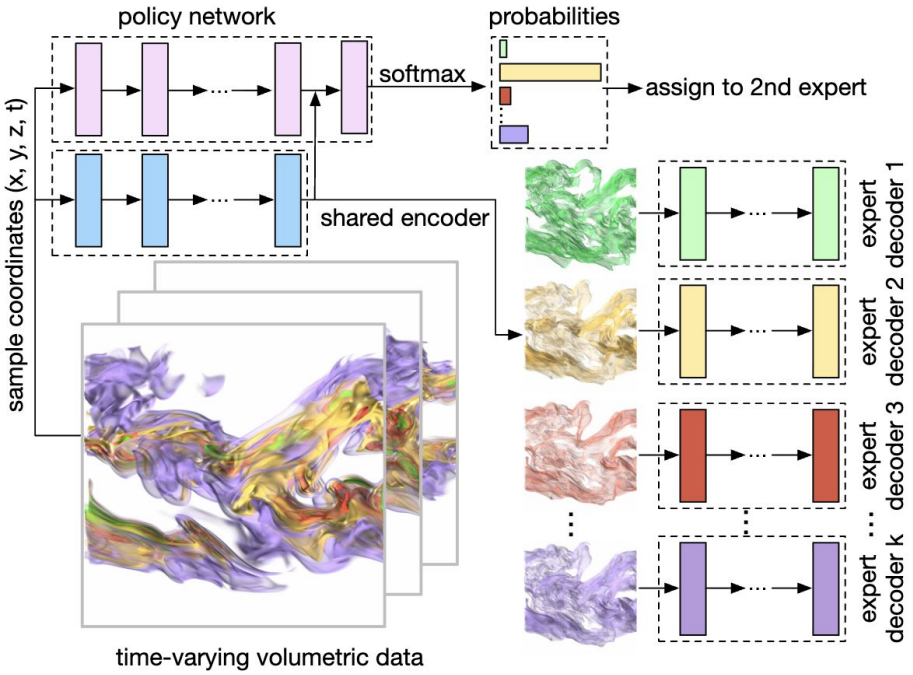


INRs beyond single volumes

Time-Varying Data
e.g., CoordNet (Han et al. 2023) and MoE-INR (Han et al. 2026)

Multivariate Data
e.g., STRS-INR (Tang et al., 2024), SWAP-INR (Son et al., 2026)

Ensemble/cross-domain Data
e.g., EVOLVE (Tang et al. 2026)



MoE-INR (Han et al. 2026)

INRs vs Compressors?

There is no simple answer

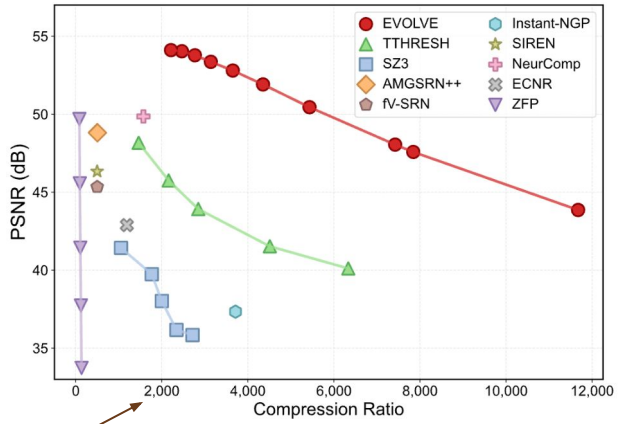
There is always a trade-off...



Credit: Gemini

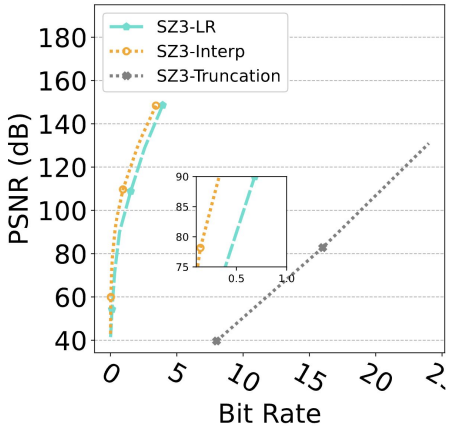
Rate Distortion Curves

- Different styles of rate-distortion curves -- Same information presented differently
  - Compression ratio vs PSNR
  - Bitrate vs PSNR
  - Distortion vs bitrate

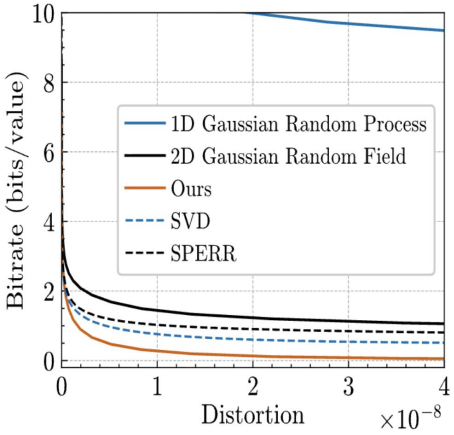


Bitrate is 0.0005

VIS community
E.g., (Tang et al. 2026)



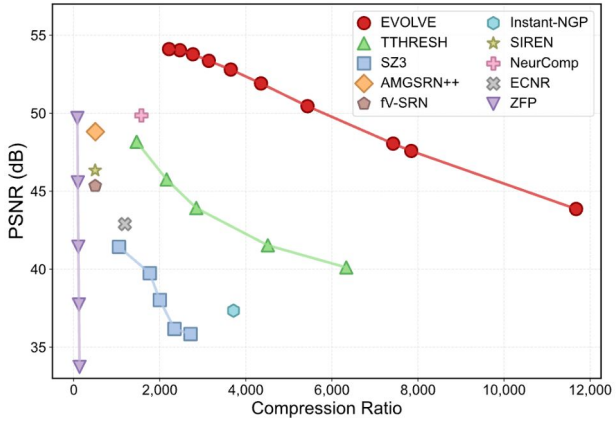
HPC community
E.g., (Liang et al. 2015)

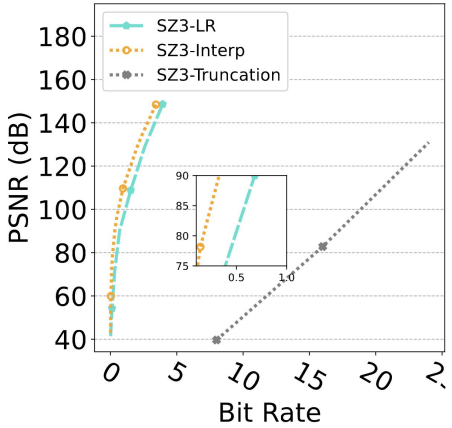


Information Theory Community
E.g., (Sinha et al. 2025)

Comparison Quality

- INRs have superior fidelity at very low bitrate
- INRs rarely show high PSNR above 55 dB
  - Lossy compressors can achieve much higher PSNR, e.g., 90 dB, which is required by scientific applications such as cosmology
- INRs have no pointwise error control, while error-bounding is important for scientific applications



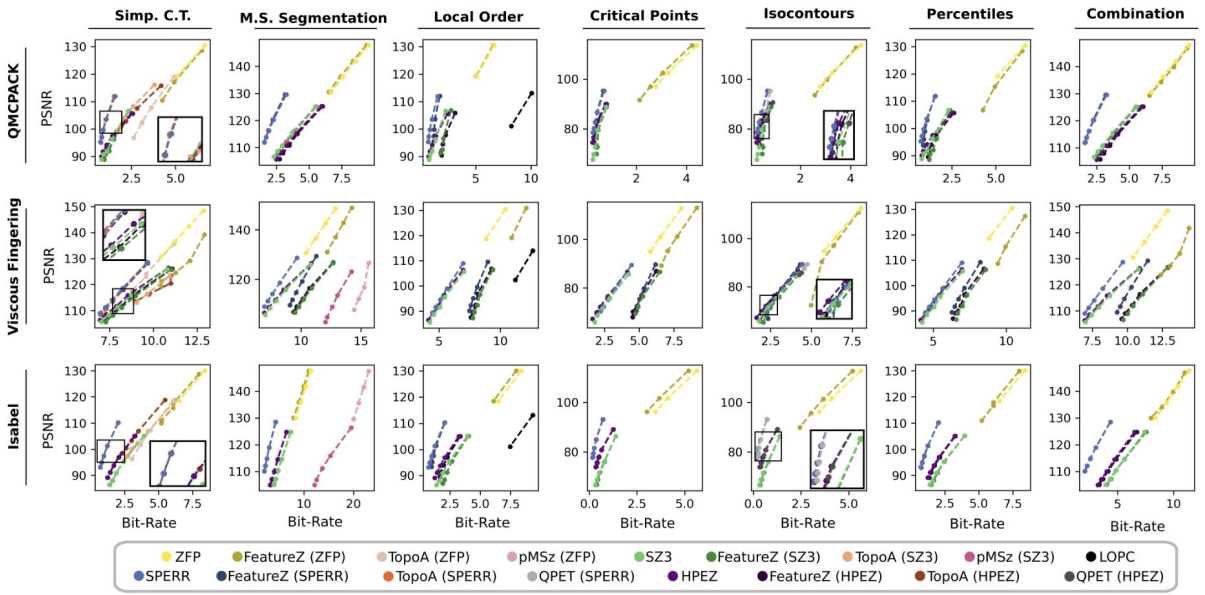


VIS community
E.g., (Tang et al. 2026)

HPC community
E.g., (Liang et al. 2015)

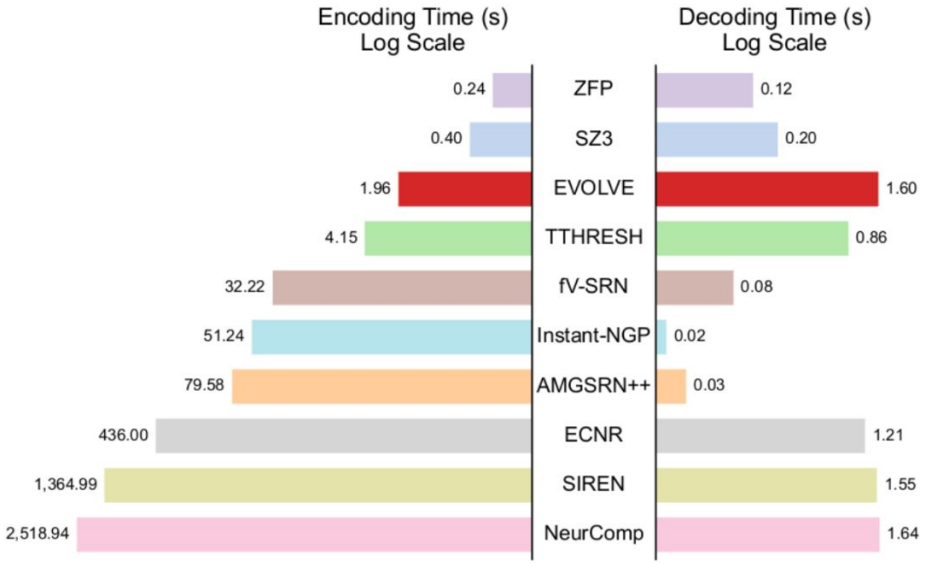
Comparison Quality on (Topological) Features

- While topology preservation considered reconstructed from INRs are more difficult to correct topologies (evidenced by FeatureZ [Gorski et al. 2026] and TopoA [Gorski et al. 2025]



Performance Gap between INRs and Compressors

- Lossy compressors can not achieve a compression throughput of 100 GB/s on a single GPU
- INRs have very low “compression” throughput because of low training speed



Tang et al. 2026

Summary: Compression vs Neural Representation

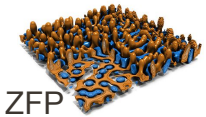
Error-bounded lossy compression

- Pointwise error control
- High compression throughput (recently achieving 100 GB/s on a single GPU)
- High decompression throughput, too
- High fidelity with low error bounds

Neural representation

- No pointwise error control
- Low “compression” throughput because of low training speed
- High “decompression” throughput with neural inference
- Generally higher fidelity with low bitrate scenarios

SZ



ZFP

MGARD

